

# Polina Petrov

Ph.D. Candidate ◇ Astrophysics

✉ [polina.petrov@vanderbilt.edu](mailto:polina.petrov@vanderbilt.edu) [polinapetrov.github.io](https://polinapetrov.github.io)

## Education

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|--------------------------------------------------------|--------------------|
| <b>Vanderbilt University</b> , Nashville, TN           | Aug 2021 - Present |
| Ph.D. in Astrophysics                                  |                    |
| Advisor: Dr. Stephen R. Taylor                         |                    |
| <b>Carnegie Mellon University</b> , Pittsburgh, PA     | 2021               |
| B.S. in Physics, Astrophysics Concentration            |                    |
| Mellon College of Science Honors and University Honors |                    |

## Honors & Awards

|             |                                                                            |           |
|-------------|----------------------------------------------------------------------------|-----------|
| 2025        | McMinn Summer Research Award, Physics and Astronomy, Vanderbilt University | \$1000    |
| 2024        | Graduate School Conference Travel Grant, Vanderbilt University             | \$1000    |
| 2023        | Travel Award, APS DGRAV                                                    | \$1000    |
| 2023 - 2026 | FINESST Grant, NASA                                                        | \$150,000 |
| 2021 - 2026 | University Graduate Fellowship, Vanderbilt University                      |           |

## Teaching Experience

### Mentoring

|                                                   |                     |
|---------------------------------------------------|---------------------|
| Tejas Mahesh, Vanderbilt University undergraduate | June 2025 - Present |
| Celine Mang, Vanderbilt University undergraduate  | Spring 2024         |

### Workshops

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|--------------------------------------------------------------------------------------------------------------------|-----------|
| <i>Individual Supermassive Black Hole Binaries and Continuous Waves</i> , NANOGrav Student Workshop, Ann Arbor, MI | Oct 2024  |
| <i>Continuous Waves</i> , VIPER PTA Summer School, Vanderbilt University                                           | July 2024 |

### Graduate Teaching Assistant, Vanderbilt University

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|---------------------------------------------------------|-------------|
| ASTR 1010L: Introductory Nighttime Astronomy Laboratory | 2021 - 2023 |
| ASTR 1020L: Introductory Daytime Astronomy Laboratory   | Fall 2022   |

## Publications

### First Author

- P. Petrov**, L. Schult, S. R. Taylor, N. Pol, N. Laal, M. Charisi, C. P. Ma, "*Expectations for the first supermassive black-hole binary resolved by PTAs II: Milestones for binary characterization*" (2025)  
[arXiv:2510.01316](https://arxiv.org/abs/2510.01316)
- P. Petrov**, S. R. Taylor, M. Charisi, C. P. Ma, "*Identifying the Host Galaxies of Supermassive Black Hole Binaries Found by Pulsar Timing Arrays*", *ApJ*, 976, 129 (2024)  
[10.3847/1538-4357/ad7b14](https://arxiv.org/abs/10.3847/1538-4357/ad7b14)
- P. Petrov**, L. P. Singer, M. W. Coughlin, [and 7 others], "*Data-driven expectations for electro-magnetic counterpart searches based on LIGO/Virgo public alerts*", *ApJ*, 924, 54 (2022)  
[10.3847/1538-4357/ac366d](https://arxiv.org/abs/10.3847/1538-4357/ac366d)

## Short Author

M. Charisi, S. R. Taylor, J. Runnoe, C. A. Witt, and **P. Petrov**, "*Multi-messenger Analysis of Supermassive Black Hole Binaries: The Joint-likelihood Approach*" (2025)  
[arXiv:2510.08683](#)

L. Schult, **P. Petrov**, S. R. Taylor, N. Pol, N. Laal, M. Charisi, C. P. Ma, "*Expectations for the first supermassive black-hole binary resolved by PTAs I: Model efficacy*" (2025)  
[arXiv:2510.01317](#)

B. Bécsy, N. J. Cornish, **P. Petrov**, X. Siemens, S. R. Taylor, S. J. Vigeland, and C. A. Witt, "*Towards robust gravitational wave detections from individual supermassive black hole binaries*", CQG, 42, 17 (2025)  
[10.1088/1361-6382/adfd36](#)

## Collaboration Publications

The NANOGrav Collaboration, "*Inferring Mbh-Mbulge Evolution from the Gravitational Wave Background*" (2025)  
[arXiv:2508.18126](#)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Targeted Searches for Supermassive Black Hole Binaries*" (2025)  
[arXiv:2508.16534](#)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Search for Gravitational-wave Memory*", ApJ, 987, 5 (2025)  
[10.3847/1538-4357/add874](#)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Harmonic Analysis of the Pulsar Angular Correlations*", ApJ, 985, 99 (2025)  
[10.3847/1538-4357/adc997](#)

The NANOGrav Collaboration, "*Galaxy Tomography with the Gravitational Wave Background from Supermassive Black Hole Binaries*" (2024)  
[arXiv:2411.05906](#)

The NANOGrav Collaboration, "*Erratum: 'The NANOGrav 15 yr Data Set: Search for Signals from New Physics' (2023, ApJL 951 L11)*", ApJL, 971, L27 (2024)  
[10.3847/2041-8213/ad68fc](#)

The IPTA Collaboration, "*Comparing Recent Pulsar Timing Array Results on the Nanohertz Stochastic Gravitational-wave Background*", ApJ, 966, 105 (2024)  
[10.3847/1538-4357/ad36be](#)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Constraints on Supermassive Black Hole Binaries from the Gravitational Wave Background*", ApJL, 952, L37 (2023)  
[10.3847/2041-8213/ace18b](#)

The NANOGrav Collaboration, "*The NANOGrav 15 yr Data Set: Search for Signals from New Physics*", ApJL, 951, L11 (2023)  
[10.3847/2041-8213/acdc91](#)

The NANOGrav Collaboration [including **P. Petrov** in analysis team], "*The NANOGrav 15 yr Data Set: Bayesian Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries*", ApJL, 951, L50 (2023)  
[10.3847/2041-8213/ace18a](#)

The NANOGrav Collaboration [including **P. Petrov** in analysis team], "*The NANOGrav 15 yr Data Set: Evidence for a Gravitational-Wave Background*", ApJL, 951, L8 (2023)  
[10.3847/2041-8213/acdac6](#)

## Presentations

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### Detection Milestones for Individual Binaries Detected by PTAs

IPTA Science Meeting, California Institute of Technology, Pasadena, CA June 2025

### Identifying Host Galaxies of Supermassive Black Hole Binaries Found by PTAs

The Era of Binary Supermassive Black Holes Meeting, Aspen, CO Feb 2025

Astronomy Department Lunch Talk, University of California, Berkeley, CA Dec 2024

IPTA Science Meeting, Sexten, Italy June 2024

### Optimizing Host Galaxy Identification of Individual Supermassive Black Hole Binaries

NANOGrav Spring Meeting, Virtual Mar 2024

AAS 243, New Orleans, LA Jan 2024

NANOGrav Fall Meeting, University of British Columbia, Vancouver, BC Oct 2023

### Mapping the Host Galaxies of Nanohertz Gravitational Wave Sources

IPTA Science Meeting, Virtual June 2023

APS April Meeting, Minneapolis, MN Apr 2023

NANOGrav Fall Meeting, University of Wisconsin-Milwaukee, Milwaukee, WI Oct 2022

GMT Meeting: Black Holes at All Scales, Sedona, AZ (Poster) Sept 2022

VIPER PTA Summer School, Nashville, TN July 2022

### Astronomy Journal Research Club, Vanderbilt University

Multi-messenger Analysis of Supermassive Black Hole Binaries: The Joint-likelihood Approach Oct 2025

Identifying Host Galaxies of Supermassive Black Hole Binaries Found by PTAs Feb 2025

The NANOGrav 15-year Data Set: Bayesian Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries Jan 2024

The NANOGrav 12.5-year Data Set: Bayesian Limits on Gravitational Waves from Individual Supermassive Black Hole Binaries Jan 2023

## Service & Outreach

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Astronomy on Tap organizer and social media coordinator, Nashville, TN 2023 - Present

Adventure Science Center Way Late Play Date, Vanderbilt Astronomy Outreach May 2025

NANOGrav Student Workshop Organizing Committee, Ann Arbor, MI Oct 2024

VIPER PTA Summer School organizer, Vanderbilt University July 2024, 2022

Radio interview on WPLN News, Nashville, TN July 2023

Adventure Science Center Astronomy Day, Vanderbilt Astronomy Outreach April 2023

Physics Concepts mentor, Carnegie Mellon University 2018

Buhl Planetarium volunteer, Pittsburgh, PA 2016 - 2018

## Skills

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**Programming** — Python, Bash, High-Performance Computing (SLURM)

**Scientific Software** — enterprise-pulsar, libstempo, QuickCW, ligo.skymap

**Other** — LaTeX, HTML, Github Pages, Microsoft Office, Google Docs, Adobe Illustrator

## Professional & Academic Affiliations

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Full Member, North American Nanohertz Observatory for Gravitational waves (NANOGrav)

Member, International Pulsar Timing Array (IPTA)

Member, American Physical Society

Member, American Astronomical Society